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Assignment

Power Electronic 2

120W Boost Converter Design

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1 Overview Of Power Electronic

1.1 What's DC/DC Converter?

A DC-DC converter is an electronic circuit that converts a source of direct current (DC) from one voltage level to another. Unlike transformers, which are designed to work with alternating current (AC), DC-DC converters are specifically tailored to handle DC power. The primary purpose of a DC-DC converter is to regulate and transform voltage levels to meet the requirements of various electronic devices. For example, step-down converters reduce voltage (for example, from 12V to 5V), while step-up converters increase voltage (e.g., from 3V to 12V).

There are several types of DC-DC converters, including:

- **Buck Converter** : Steps down voltage.
- **Boost Converter** : Steps up voltage.
- **Buck-Boost Converter** : Can both step up and step down voltage.
- **Flyback Converter** : Used for isolation and multiple outputs.

DC-DC converters are used in a wide range of applications, including consumer electronics (smartphones, laptops, and tablets), automotive systems (electric vehicles, infotainment systems, and lighting), renewable energy (solar panels, battery storage systems, and wind turbines) and industrial equipment (motor drives, robotics, and control systems).

1.2 Importance of Buck Converters

Boost converters play a vital role in modern power electronics due to their ability to efficiently step up voltage levels. Their importance lies in their capacity to generate higher output voltages from low-voltage sources, ensuring stable and regulated power delivery even under fluctuating input conditions. They are highly efficient, minimizing energy loss and heat generation, which is critical for optimizing system performance. Additionally, their compact design and wide input voltage range make them adaptable to diverse operating environments. Boost converters help increase voltage in different systems. They provide steady voltage control and support modern technologies, making them effective, affordable, and easy to expand.

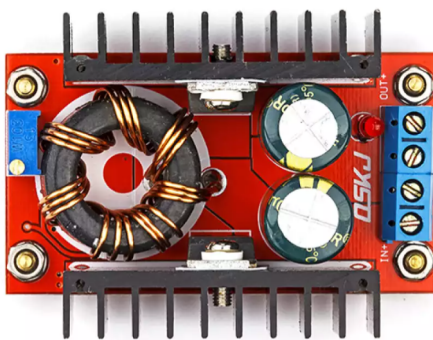


Figure 1: *150W Boost Converter*

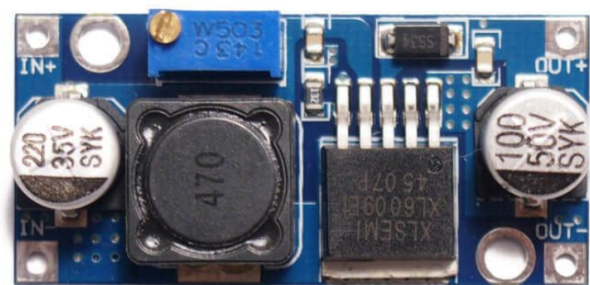


Figure 2: *5V-35V Boost Converter*

2 Fundamental Of Buck Converter

2.1 Basic Operation and Working Principle

is a type of DC-DC converter that steps up a lower input voltage to a higher output voltage. It operates on the principle of **energy storage and transfer** using an inductor, a switch, and a diode. The operation alternates between two main phases: **Switch ON (Energy Storage)** and **Switch OFF (Energy Transfer)**.

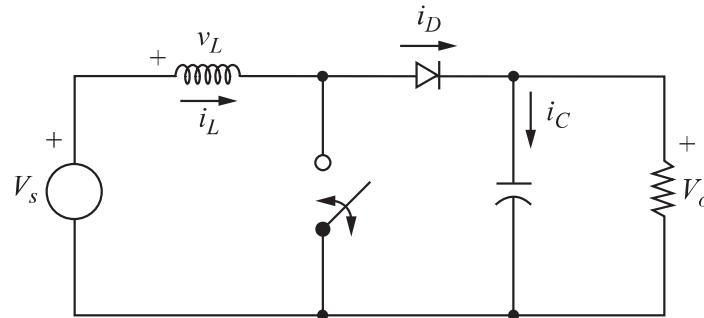


Figure 3: *The boost converter*

Switch ON (Energy Storage Phase)

When the switch (typically a MOSFET) is closed (ON), current flows through the inductor and the switch. During this phase:

- The inductor stores energy in its magnetic field.
- The diode is reverse-biased, so no current flows to the load or the output capacitor.
- The voltage across the inductor is given by:

$$V_L = V_{in}$$

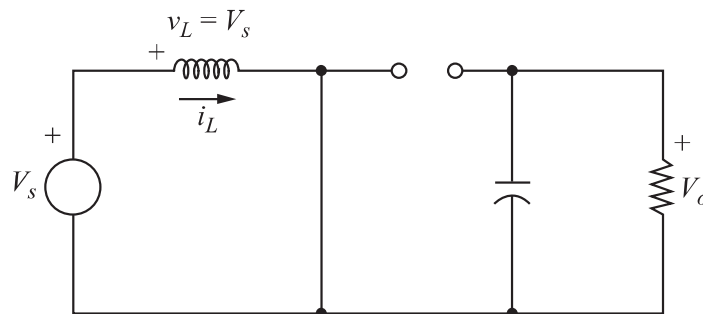


Figure 4: *Equivalent circuit for the switch closed*

Switch OFF (Energy Transfer Phase)

When the switch is opened (OFF):

- The inductor resists the sudden drop in current by inducing a voltage that opposes the change.
- This induced voltage forward-biases the diode, allowing current to flow from the inductor to the load and the output capacitor.
- The output capacitor smooths out the voltage, providing a stable DC output.
- The voltage across the inductor during this phase is:

$$V_L = V_{out} - V_{in}$$

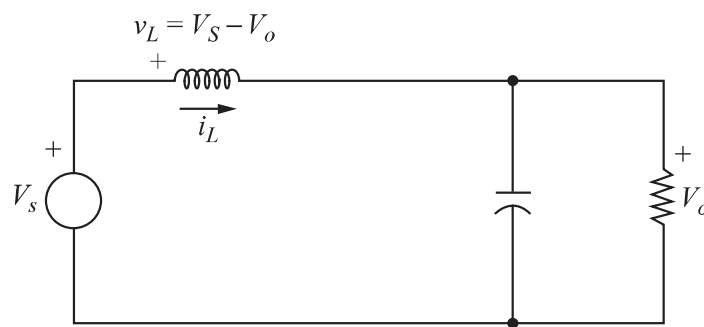


Figure 5: *Equivalent circuit for the switch open*

2.2 Key Components

The basic boost converter circuit consists of four key components:

1. Inductor (L)

- **Function:** Stores energy in the form of a magnetic field when the switch is ON and releases it to the load when the switch is OFF.
- **Role:** Smooths out the current ripple and ensures continuous energy transfer to the load.
- **Selection Criteria:** The inductor value determines the amount of energy stored and affects the ripple current. Larger inductors reduce ripple but increase size and cost.

2. Switch (S)

- **Function:** Controls the flow of current through the inductor. Typically implemented using a MOSFET or transistor.
- **Role:** The switch alternates between ON and OFF states at a high frequency (controlled by a PWM signal).
- **Selection Criteria:** The switch must handle the maximum input voltage and current with minimal power loss (low $R_{ds(on)}$ for MOSFETs).

3. Diode (D)

- **Function:** Acts as a freewheeling diode, providing a path for the inductor current when the switch is OFF.
- **Role:** Prevents the inductor from discharging back into the switch and ensures continuous current flow to the load.
- **Selection Criteria:** A Schottky diode is often used due to its low forward voltage drop and fast switching speed, which minimizes power loss.

4. Capacitor (C)

- **Function:** Smooths out the output voltage and reduces voltage ripple.
- **Role:** Stores charge during the OFF phase and releases it during the ON phase, ensuring a stable DC output.
- **Selection Criteria:** The capacitor value affects the output voltage ripple. Larger capacitors reduce ripple but may increase cost and size.

2.3 Duty Cycle, Input/Output Voltage Relationship, and Efficiency

1. Duty Cycle (D)

- **Definition:** The duty cycle is the fraction of time the switch is ON during a switching period.
- **Formula:**

$$D = \frac{T_{on}}{T_{on} + T_{off}} = \frac{T_{on}}{T}$$

Where:

- T_{on} : Time the switch is ON
- T_{off} : Time the switch is OFF
- T : Total switching period ($T = T_{on} + T_{off}$)

2. Input/Output Voltage Relationship

The relationship between the input voltage (V_{in}) and the output voltage (V_{out}) in a boost converter is determined by the duty cycle (D).

- **Formula:**

$$V_{out} = \frac{V_{in}}{1 - D}$$

Rearranging for D :

$$D = 1 - \frac{V_{in}}{V_{out}}$$

3. Efficiency

- **Definition:** Efficiency is the ratio of output power to input power, expressed as a percentage.
- **Formula:**

$$\text{Efficiency}(\eta) = \frac{P_{out}}{P_{in}} \times 100\%$$

Where:

- $P_{out} = V_{out} \cdot I_{out}$
- $P_{in} = V_{in} \cdot I_{in}$

3 Mathematical Modeling

3.1 Derivation of the Voltage Conversion Equation

The goal of this section is to derive the relationship between the input voltage (V_{in}) and the output voltage (V_{out}) in a boost converter. The derivation involves analyzing the two phases of operation (Switch ON and Switch OFF) and applying the principle of energy balance.

Step 1: Understanding the Two Phases of Operation

A boost converter operates in two distinct phases:

1. **Switch ON (Energy Storage Phase):** The switch is closed, and current flows through the inductor, storing energy in its magnetic field.
2. **Switch OFF (Energy Transfer Phase):** The switch is open, and the inductor releases its stored energy to the load through the diode.

We will analyze these two phases to derive the voltage conversion equation.

Step 2: Analyzing the Switch ON Phase

During the **ON phase**, the switch is closed, and the inductor is directly connected to the input voltage source (V_{in}). The voltage across the inductor is:

$$V_L = V_{in}$$

The current through the inductor increases linearly with time because the inductor stores energy. The rate of change of inductor current is given by:

$$\frac{dI_L}{dt} = \frac{V_L}{L} = \frac{V_{in}}{L}$$

where: - L : Inductance of the inductor - t : Time

The increase in inductor current during the ON phase is proportional to the duration of the ON phase (T_{on}):

$$\Delta I_{on} = \frac{V_{in}}{L} \cdot T_{on}$$

Step 3: Analyzing the Switch OFF Phase

During the **OFF phase**, the switch is open, and the inductor tries to maintain the current flow (due to its stored energy). This causes the inductor to induce a voltage that adds to the input voltage. The voltage across the inductor during this phase is:

$$V_L = V_{out} - V_{in}$$

The current through the inductor decreases linearly with time. The rate of change of inductor current is given by:

$$\frac{dI_L}{dt} = \frac{V_L}{L} = \frac{V_{out} - V_{in}}{L}$$

The decrease in inductor current during the OFF phase is proportional to the duration of the OFF phase (T_{off}):

$$\Delta I_{off} = \frac{V_{out} - V_{in}}{L} \cdot T_{off}$$

Step 4: Applying the Principle of Energy Balance

To ensure continuous operation, the average current through the inductor over one switching cycle must remain constant. This means that the increase in inductor current during the ON phase must equal the decrease in inductor current during the OFF phase:

$$\Delta I_{on} = \Delta I_{off}$$

Substitute the expressions for ΔI_{on} and ΔI_{off} :

$$\frac{V_{in}}{L} \cdot T_{on} = \frac{V_{out} - V_{in}}{L} \cdot T_{off}$$

Cancel out L (since it is the same in both phases):

$$V_{in} \cdot T_{on} = (V_{out} - V_{in}) \cdot T_{off}$$

Step 5: Relating T_{on} and T_{off} to the Duty Cycle

From the definition of the duty cycle:

$$D = \frac{T_{on}}{T}, \quad T_{off} = T - T_{on} = T \cdot (1 - D)$$

where: - T : Total switching period ($T = T_{on} + T_{off}$)

Substitute $T_{on} = D \cdot T$ and $T_{off} = (1 - D) \cdot T$ into the equation:

$$V_{in} \cdot (D \cdot T) = (V_{out} - V_{in}) \cdot ((1 - D) \cdot T)$$

Cancel out T (since it is the same on both sides):

$$V_{in} \cdot D = (V_{out} - V_{in}) \cdot (1 - D)$$

Expand the right-hand side:

$$V_{in} \cdot D = V_{out} \cdot (1 - D) - V_{in} \cdot (1 - D)$$

Simplify:

$$V_{in} \cdot D = V_{out} - V_{out} \cdot D - V_{in} + V_{in} \cdot D$$

Cancel out $V_{in} \cdot D$ on both sides:

$$0 = V_{out} - V_{out} \cdot D - V_{in}$$

Factor out V_{out} :

$$V_{out} \cdot (1 - D) = V_{in}$$

Solve for V_{out} :

$$V_{out} = \frac{V_{in}}{1 - D}$$

3.2 Mode of Operation

Continuous Conduction Mode (CCM)

In **CCM**, the inductor current never drops to zero during the switching cycle. This means that energy is continuously transferred from the input to the output through the inductor. CCM is typically used in high-power applications where the load current is relatively high.

- The inductor current has a **triangular waveform** with a DC offset.
- The converter operates in steady-state, meaning the average inductor current remains constant over time.
- The ripple current is small compared to the average inductor current.

Key Characteristics

- The inductor current is always positive ($I_L > 0$).
- The output voltage depends only on the duty cycle (D) and input voltage (V_{in}).
- The ripple current is small compared to the average inductor current.

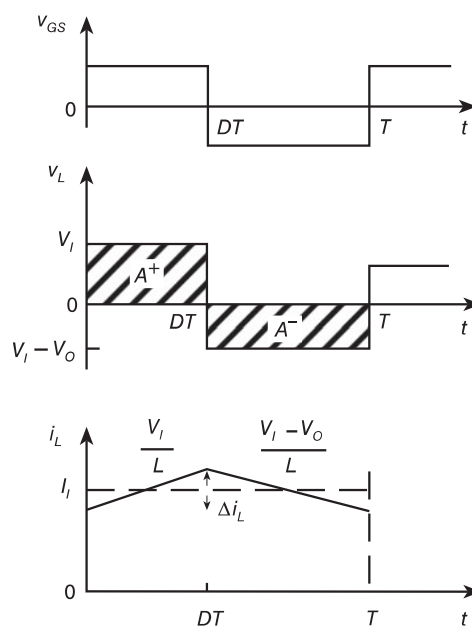


Figure 6: *CCM Operation Mode*

Discontinuous Conduction Mode (DCM)

DCM Operation Mode

In **DCM**, the inductor current drops to zero before the next switching cycle begins. This mode occurs when the load current is low, and the inductor cannot maintain continuous current flow. DCM is typically used in low-power applications.

- The inductor current waveform has a **triangular shape** but drops to zero during part of the switching cycle.
- The output voltage depends not only on the duty cycle (D) but also on the load current and inductor value.

Key Characteristics

- The inductor current is zero for part of the switching cycle ($I_L = 0$).
- The output voltage is less stable compared to CCM.
- Efficiency may drop at very light loads.

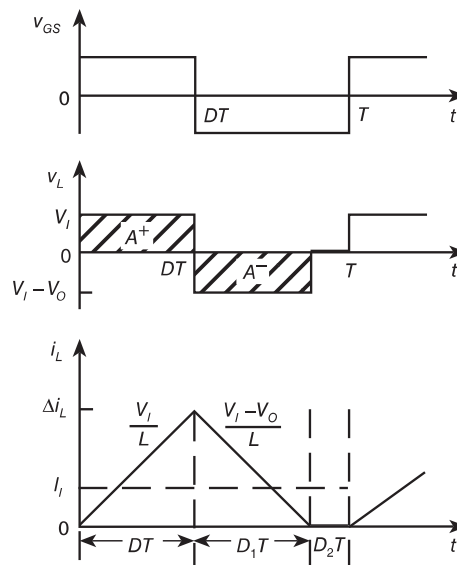


Figure 7: *DCM Operation Mode*

4 Design and Implementation

4.1 Power Stage Calculation and Component Selection

Design of a 120W Boost Converter Operating in Continuous Conduction Mode (CCM) with the following specifications:

Parameter	Value
Output Voltage (V_{out})	24 V
Input Voltage (V_{in})	12 V
Output Power (P_{out})	120 W
Switching Frequency (f_{sw})	100 kHz
Ripple Current Rated	30%
Ripple Voltage Rated	1%

Table 1: Parameters Needed for Power Stage Calculation

Power Stage Calculations

1. Duty Cycle (D)

$$D = 1 - \frac{V_{in}}{V_{out}} = 1 - \frac{12}{24} = 0.5 \quad (50\%)$$

2. Output Current (I_{out})

$$I_{out} = \frac{P_{out}}{V_{out}} = \frac{120}{24} = 5 \text{ A}$$

3. Output Resistance (R)

$$R_{out} = \frac{V_{out}^2}{P_{out}} = \frac{24^2}{120} = 4.8 \Omega$$

4. Inductor Ripple Current (ΔI_L)

- Average Inductor Current:

$$I_{L_{avg}} = \frac{V_{out} \cdot I_{out}}{V_{in}} = \frac{24 \cdot 5}{12} = 10 \text{ A}$$

- Inductor Ripple Current:

$$\Delta I_L = 0.3 \cdot I_{L_{avg}} = 0.3 \cdot 10 = 3 \text{ A (peak-to-peak)}$$

- Minimum Inductor Current:

$$I_{L_{max}} = I_{L_{avg}} + \frac{\Delta I_L}{2} = 10 - \frac{3}{2} = 8.5 \text{ A}$$

- Maximum Inductor Current:

$$I_{L_{min}} = I_{L_{avg}} - \frac{\Delta I_L}{2} = 10 + \frac{3}{2} = 11.5 \text{ A}$$

5. Ripple Voltage (ΔV_{out})

$$\Delta V_{out} = 0.01 \cdot V_{out} = 0.24 \text{ V}$$

6. Output Capacitor (C)

$$C = \frac{I_{out} \cdot D}{F_s \cdot \Delta V_{out}} = \frac{5 \cdot 0.5}{100 \times 10^3 \cdot 0.24} \approx 104 \mu\text{F}$$

7. Inductor (L)

$$L = \frac{V_{in} \cdot D}{F_s \cdot \Delta I_L} = \frac{12 \cdot 0.5}{100 \times 10^3 \cdot 3} = 20 \mu\text{H}$$

8. MOSFET Selection



Table 2: IRF540N MOSFET

Parameter	Value	Test Conditions
V_{DS}	100 V	Absolute Maximum Rating
I_D	14 A	$T_C = 25^\circ\text{C}$
	10 A	$T_C = 100^\circ\text{C}$
$R_{DS(on)}$	0.16Ω	$V_{GS} = 10 \text{ V}, I_D = 8.4 \text{ A}$

9. Diode Selection



Table 3: 15SQ045 Schottky Diode

Parameter	Value	Test Conditions
V_{RRM}	100 V	Maximum repetitive rating
I_F	5 A	Continuous, $T_A = 25^\circ\text{C}$

11. Inductor Design

• Inductor Core Selection Using Kg Method

Table 4: Input Parameters for Ferrite Core Design

Symbol	Value	Description
L	$20 \mu\text{H}$	Inductance
I_o	10 A	DC Current
ΔI	3 A	AC Ripple Current
P_o	120 W	Output Power
α	1%	Regulation
f_s	100 kHz	Switching Frequency
B_m	0.25 T	Operating Flux Density
K_u	0.4	Window Utilization
T_r	25°C	Temperature Rise Goal

- Peak Current

$$I_{pk} = I_o + \frac{\Delta I}{2} = 10 + \frac{3}{2} = \boxed{11.5 \text{ A}}$$

- Energy-Handling Capability

$$E = \frac{1}{2} L I_{pk}^2 = \frac{1}{2} \cdot 20 \times 10^{-6} \cdot (11.5)^2 \approx \boxed{0.00132 \text{ W-s}}$$

- Electrical Condition Coefficient

$$K_e = 0.145 \cdot P_o \cdot B_m^2 \cdot 10^{-4} = 0.145 \cdot 120 \cdot (0.25)^2 \cdot 10^{-4} \approx \boxed{1.075 \times 10^{-4}}$$

- Core Geometry Coefficient

$$K_g = \frac{E^2}{K_e \cdot \alpha} = \frac{(0.00132)^2}{1.075 \times 10^{-4} \cdot 1} \approx \boxed{0.0161 \text{ cm}^5}$$

PQ, Ferrite Cores (TDK)											
Part No.	W _{tcu} grams	W _{tfe} grams	MLT cm	MPL cm	W _a	A _c	W _a	A _p	K _g	A _t	*AL
					A _c	cm ²	cm ²	cm ⁴	cm ⁵	cm ²	mh/1K
PQ20/16	7.4	13.0	4.4	3.74	0.765	0.620	0.474	0.294	0.0167	16.9	1617
PQ20/20	10.4	15.0	4.4	4.54	1.061	0.620	0.658	0.408	0.0227	19.7	1313
PQ26/20	31.0	31.0	5.6	4.63	0.508	1.190	0.604	0.718	0.0613	28.4	2571
PQ26/25	17.0	36.0	5.7	5.55	0.716	1.180	0.845	0.997	0.0832	32.6	2187
PQ32/20	18.9	42.0	6.6	5.55	0.475	1.700	0.808	1.373	0.1417	36.3	3046
PQ32/30	35.5	55.0	6.7	7.46	0.929	1.610	1.496	2.409	0.2326	46.9	2142
PQ35/35	59.0	73.0	7.5	8.79	1.126	1.960	2.206	4.324	0.4510	60.7	2025
PQ40/40	97.2	95.0	8.4	10.20	1.622	2.010	3.260	6.552	0.6280	77.1	1792
PQ50/50	158.5	195.0	10.3	11.30	1.321	3.280	4.332	14.209	1.8120	113.9	2800

*This AL value has been normalized for a permeability of 1K. For a close approximation of AL for other values of permeability, multiply this AL value by the new permeability in kilo-perm. If the new permeability is 2500, then use 2.5.

Figure 8: PQ Core Parameter Table List

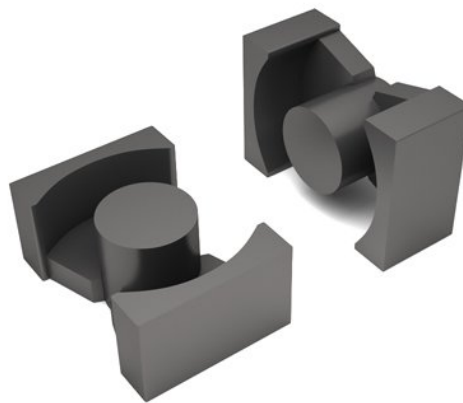


Figure 9: PQ20/20 Core Selected

Parameter	Value	Description
MPL	4.54	Magnetic Path Length (cm)
We	15	Core Weight (g)
MLT	4.4	Mean Length per Turn (cm)
A_c	0.620	Cross-Sectional Area (cm ²)
W_a	0.658	Window Area (cm ²)
A_p	0.408	Core Product (cm ⁴)
K_g	0.0227	Core Geometry (cm ⁵)

Table 5: Core Parameters for Ferrite Design

- **Current Density (J)**

$$J = \frac{2 \cdot E \cdot 10^4}{B_m \cdot A_p \cdot K_u} = \frac{2 \cdot 0.0013 \cdot 10^4}{0.25 \cdot 0.408 \cdot 0.4} \approx \boxed{637.25 \text{ A/cm}^2}$$

- **RMS Current (I_{RMS})**

$$I_{rms} = \sqrt{I_o^2 + \Delta I^2} = \sqrt{10^2 + 3^2} = \boxed{10.44 \text{ A}}$$

- **Bare Wire Area (A_w)**

$$A_w = \frac{I_{rms}}{J} = \frac{10.44}{637.25} \approx \boxed{0.0164 \text{ cm}^2}$$

- **Selected Bare Wire Area**

$$A_{ws} = 0.015 \text{ cm}^2$$

- **Effective Window Area (W_{aeff})**

- $S_3 = 0.75$ Window Effective factor for Ferrite core

$$W_{aeff} = W_a \cdot S_3 = 0.658 \cdot 0.65 \approx \boxed{0.4277 \text{ cm}^2}$$

- **Number of Turns (N)**

- $S_2 = 0.6$ Fill Factor

$$N = \frac{W_{aeff} \cdot S_2}{A_{ws}} = \frac{0.4277 \cdot 0.6}{0.015} \approx \boxed{17.11 \text{ turns}}$$

- **Required Air Gap (L_g)**

$$L_g = \frac{0.4\pi N^2 A_c \cdot 10^{-8}}{L} - \frac{MPL}{\mu_0}$$

Substituting values:

$$L_g = \left(\frac{0.4 \cdot \pi \cdot (17.11)^2 \cdot 0.620 \cdot 10^{-8}}{20 \times 10^{-6}} \right) - \left(\frac{4.54}{2500} \right) \approx \boxed{0.112 \text{ cm}}$$

4.2 Power Stage Calculation and Component Selection Using Matlab Script

Below is the MATLAB code use to calculate the Power Stage and Component Selection for the 120W Boost Converter design.

```

%%MATLAB Script for Boost Converter Design in Continuous Conduction Mode (CCM)
clc,clear;
%% Input Parameters
Vin = 12; % Input Voltage (V)
Vout = 24; % Output Voltage (V)
fsw = 100e3; % Switching Frequency (Hz)
Pout = 120; % Output Power (W)
RippleCurrentRated = 0.3; % Ripple Current Rated (30% of average inductor current)
RippleVoltageRated = 0.01; % Ripple Voltage Rated (1% of output voltage)
%% Step-by-Step Calculations
% Step 1: Calculate Duty Cycle (D)
D = 1 - (Vin / Vout);
% Step 2: Calculate Output Current (Iout)
Iout = Pout / Vout;
% Step 3: Calculate Load Resistance (Rload)
Rload = Vout / Iout;
% Step 4: Calculate Ripple Output Voltage (DeltaVout)
DeltaVout = RippleVoltageRated * Vout;
% Step 5: Calculate Inductor Value (L)
IL = Iout / (1 - D); % Average Inductor Current
DeltaIL = RippleCurrentRated * IL; % Ripple Current
L = (Vin * D) / (fsw * DeltaIL); % Inductance in Henries
% Step 6: Calculate Maximum and Minimum Inductor Currents
ILmax = IL + DeltaIL / 2; % Maximum Inductor Current
ILmin = IL - DeltaIL / 2; % Minimum Inductor Current
C = (Iout * D) / (fsw * DeltaVout); % Capacitance in Farads

```

Listing 1: MATLAB Code for 120W Boost Converter Design

4.3 Simulation On LTspice (Open Loop)

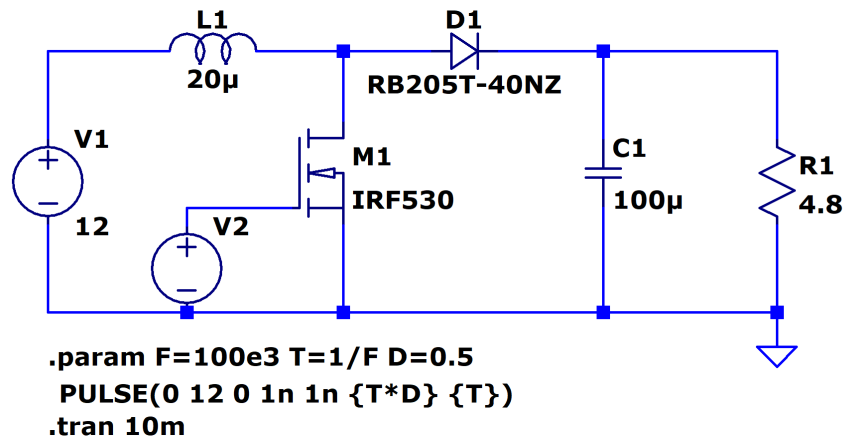


Figure 10: *Boost Converter Schematic on LTspice*

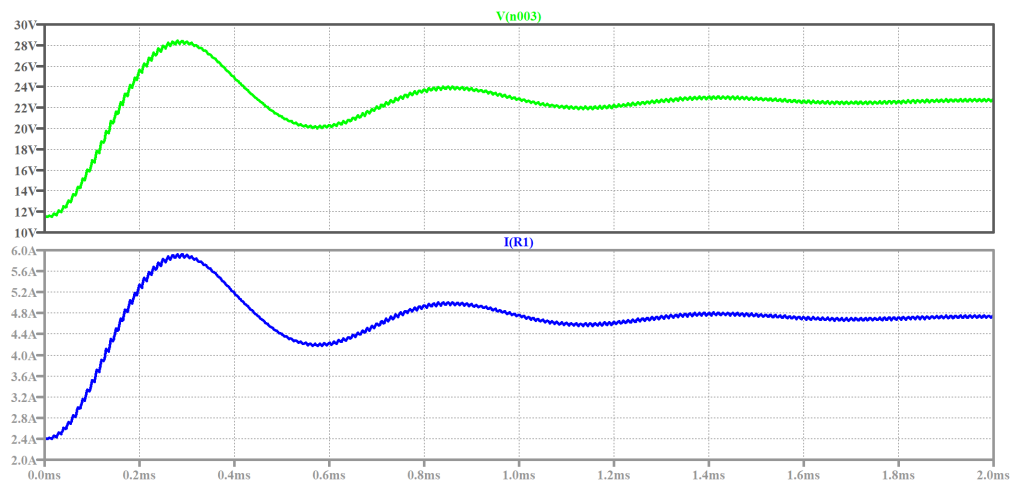


Figure 11: *Output Voltage and Output Current Waveform*

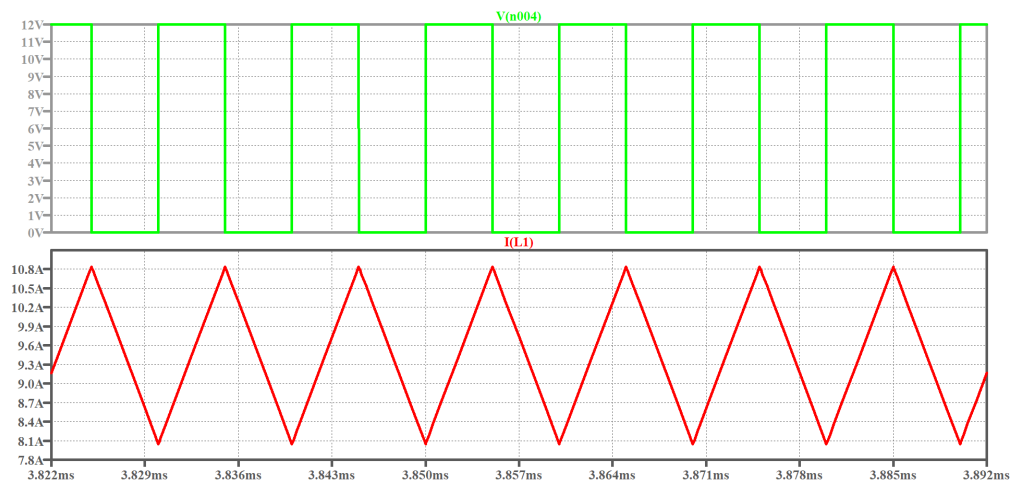


Figure 12: *Duty Cycle and Inductor Current Waveform*

4.4 Simulation On Simulink (Open Loop)

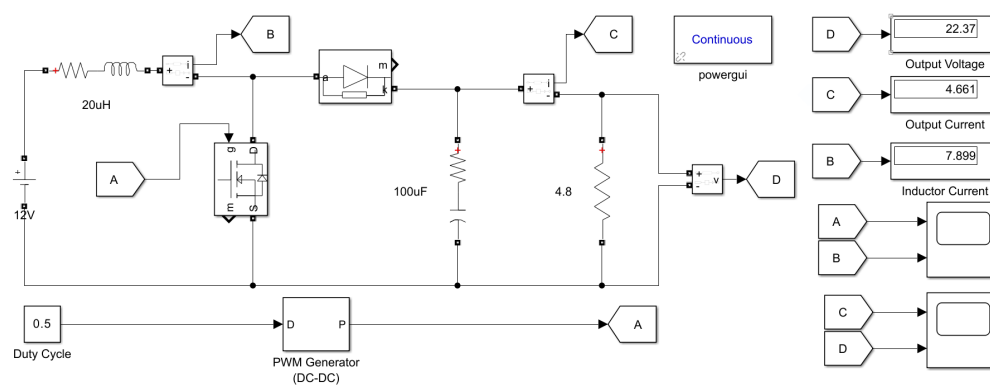


Figure 13: *Boost Converter Schematic on LTspice*

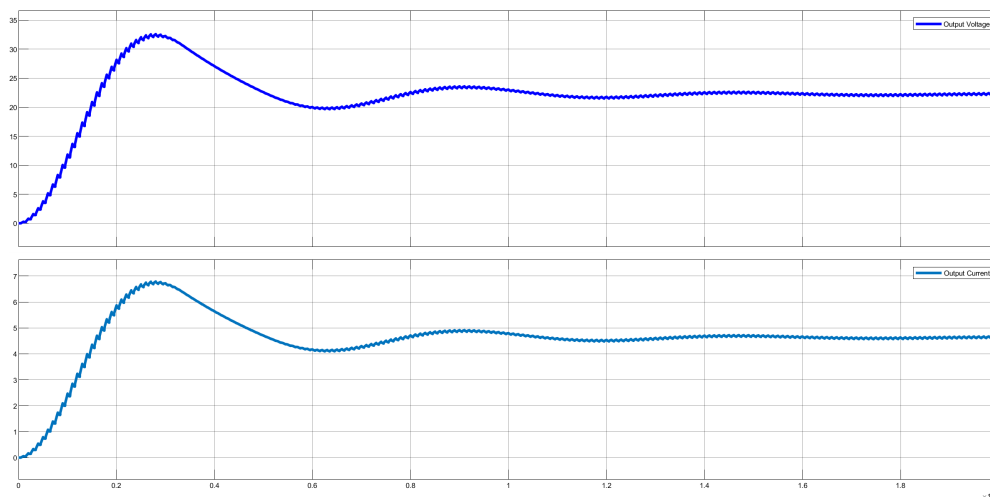


Figure 14: *Output Voltage and Output Current Waveform*

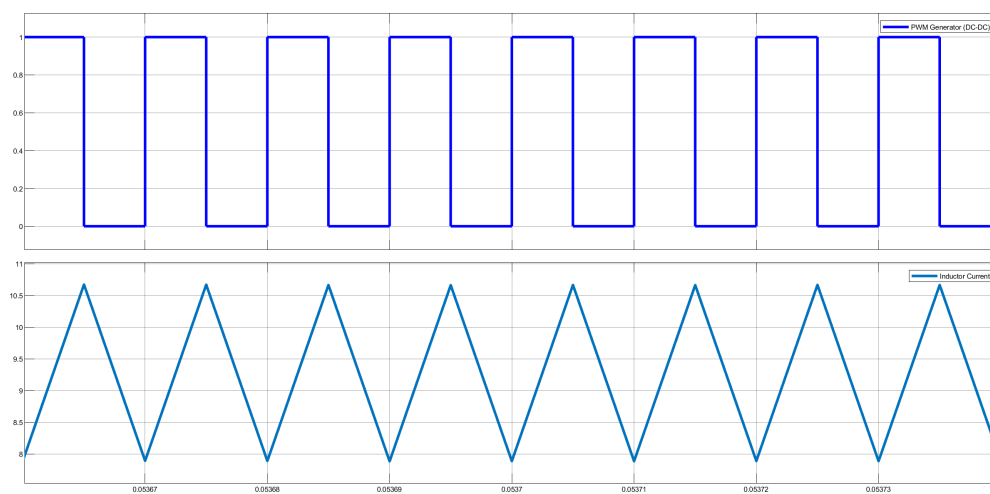


Figure 15: *Duty Cycle and Inductor Current Waveform*

4.5 Simulation on Simulink Using PI Controller (Close Loop)

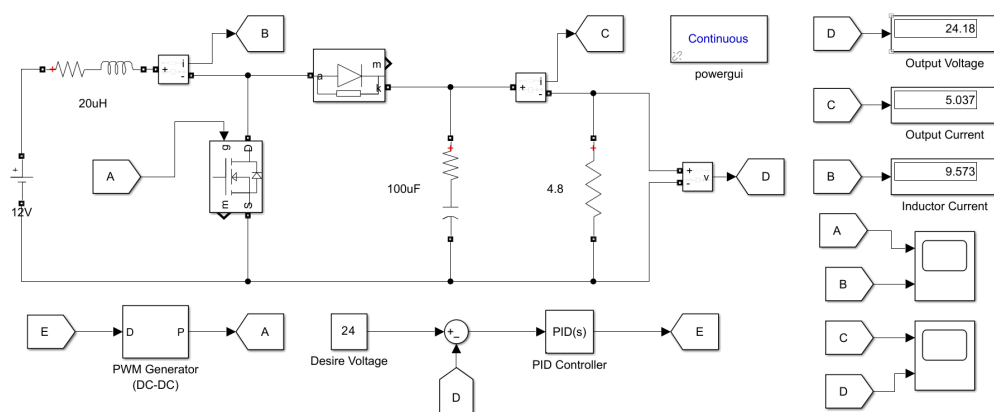


Figure 16: *Boost Converter Schematic on LTspice*

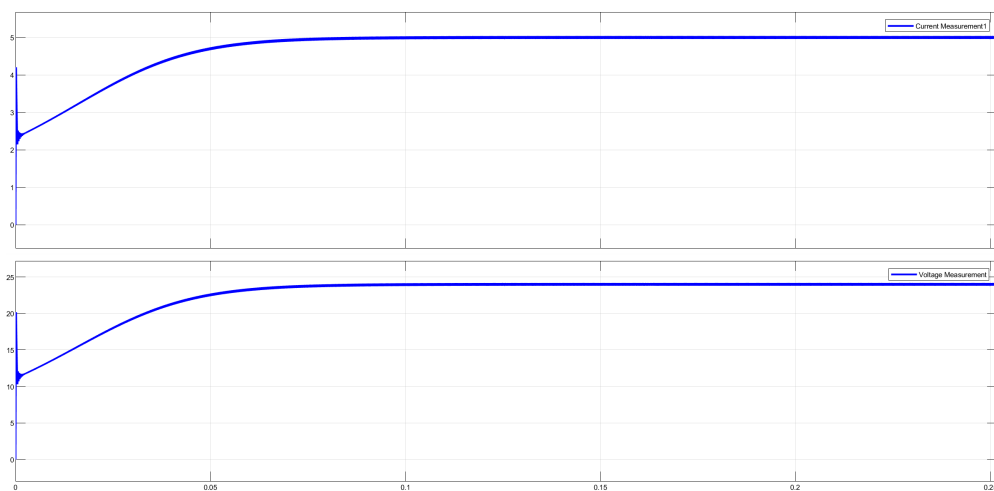


Figure 17: *Output Voltage and Output Current Waveform*

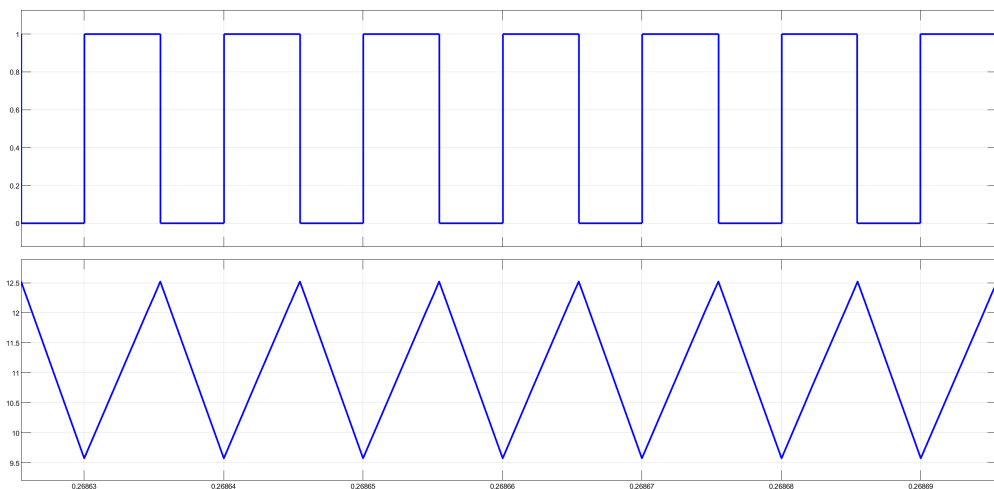


Figure 18: *Duty Cycle and Inductor Current Waveform*